

3.3.4 Alkenes

In alkenes, the high electron density of the carbon–carbon double bond leads to attack on these molecules by electrophiles. This section also covers the mechanism of addition to the double bond and introduces addition polymers, which are commercially important and have many uses in modern society.

3.3.4.1 Structure, bonding and reactivity

Content	Opportunities for skills development
Alkenes are unsaturated hydrocarbons. Bonding in alkenes involves a double covalent bond, a centre of high electron density.	

3.3.4.2 Addition reactions of alkenes

Content	Opportunities for skills development
Electrophilic addition reactions of alkenes with HBr, H₂SO₄ and Br₂ The use of bromine to test for unsaturation. The formation of major and minor products in addition reactions of unsymmetrical alkenes. Students should be able to: - outline the mechanisms for these reactions - explain the formation of major and minor products by reference to the relative stabilities of primary, secondary and tertiary carbocation intermediates.	AT d and k PS 4.1 Students could test organic compounds for unsaturation using bromine water and record their observations.

3.3.4.3 Addition polymers

Content	Opportunities for skills development
Addition polymers are formed from alkenes and substituted alkenes. The repeating unit of addition polymers. IUPAC rules for naming addition polymers. Addition polymers are unreactive. Appreciate that knowledge and understanding of the production and properties of polymers has developed over time. Typical uses of poly(chloroethene), commonly known as PVC, and how its properties can be modified using a plasticiser. Students should be able to: - draw the repeating unit from a monomer structure - draw the repeating unit from a section of the polymer chain - draw the structure of the monomer from a section of the polymer - explain why addition polymers are unreactive - explain the nature of intermolecular forces between molecules of polyalkenes.	AT k PS 1.2 Making poly(phenylethene) from phenylethene.